BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, LAHORE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2021

PHYSICS

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Objective Type (Group-I)

Time Allowed: 20 Min. | **Maximum Marks:** 17

Note: Inter, Part I, Lahore, 2021. Session 2017-2019 to 2020-2022. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. The maximum height attained by a projectile is:

(A) $vi^2\sin^2\theta/2g$
(B) $vi^2\sin^2\theta/g$
(C) $vi^2\cos^2\theta/2g$
(D) $vi^2\cos^2\theta/g$

Q2. One radian is equal to:

(A) 57.3°
(B) 67.3°
(C) 87.3°
(D) 60°

Q3. The property of bending of light around obstacles is:

(A) Reflection
(B) Refraction
(C) Diffraction
(D) Polarization

Q4. The K.E. of molecules of an ideal gas at absolute zero will be:

(A) Zero
(B) Infinite
(C) Very high
(D) Below zero

Q5. Travel time of light from sun to earth is:

(A) 1 min 20 s
(B) 8 min 20 s
(C) 10 min 20 s
(D) 5 h 20 s

Q6. The frequency range of Dog ear is:

(A) 20-20,000 Hz
(B) 60-70,000 Hz
(C) 1000-120,000 Hz
(D) 15-50,000 Hz

Q7. Unit of torque in SI unit is:

(A) $N-m^{-1}$
(B) $N^{-1}-m$
(C) $N^{-1}-m^{-1}$
(D) N−m

Q8. At resonance, the transfer of energy is:

(A) Zero
(B) Minimum
(C) Maximum
(D) Negative

Q9. The efficiency of diesel engine is about:

(A) 25-30%
(B) 35-40%
(C) 35-50%
(D) 45-60%

Q10. Kilowatt hour is the unit of:

(A) Power
(B) Work
(C) Force
(D) Momentum

Q11. The dimensions of relation mc^2 are equal to the dimensions of:

(A) Force
(B) Momentum
(C) Energy
(D) Torque

Q12. If a body of mass 10 kg is allowed to fall freely, its apparent weight becomes:

(A) Zero
(B) 89 N
(C) 9.8 N
(D) 10N

Q13. The angle of projection of a projectile for which its maximum height and horizontal range are equal to:

(A) 45°
(B) 56°
(C) 66°
(D) 76°

Q14. A ten meter high tank is full of water. A hole appears at its middle. The speed of efflux be:

(A) 5ms
(B) 9.9 ms^{-1}
(C) 8.9 ms^{-1}
(D) 5.11 ms^{-1}

Q15. The dot product $\hat{i}\cdot\hat{i}$ is equal to:

(A) 0
(B) 1
(C) -1
(D) $\hat{j}$

Q16. Beats can be heard by man when difference of frequency is not more than:

(A) 3 Hz (B) 9 Hz

(C) 10 Hz (D) 16 Hz

Q17. The least distance of distinct vision for normal eye is:

(A) 10 cm (B) 20 cm

(C) 25 cm (D) 30 cm

Essay Type (Group-I)

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

Note: Session 2017-2019 to 2020-2022

QSECTION - I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Write the dimensions of: (a) pressure. (b) density.

ii How the uncertainty in the time period of vibrating body is found?

iii Write two uses of dimensions.

iv Does dimensional analysis give any information on constant of proportionality that may appear in an algebraic expression? Explain.

v Suppose the sides of a closed polygon represent vectors arranged head to tail. What is the sum of these vectors?

vi Two vectors have unequal magnitudes. Can their sum be zero? Explain.

vii If $A = 2\hat{i} + 3\hat{j} - \hat{k}$, $B = 3\hat{i} - 2\hat{k}$, Find $A \cdot B$

viii Define range of projectile. Derive its expression.

ix State the law of conservation of linear momentum, pointing out the importance of isolated system.

x Find the angle of projection for which range of projectile is equal to four times the maximum height.

xi What is the principle of rocket propulsion?

xii Explain, how the swing is produced in a fast moving cricket ball.

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i Calculate the work done in kilo joules in lifting a mass of 10 kg (at a steady velocity) through a vertical height of 10m.

ii When a rocket re-enters the atmosphere, its nose cone becomes very hot. Where does this heat energy come from?

iii Define: (a) gravitational field. (b) conservative field.

iv Prove that $s = r\theta$ where θ is in radian.

v When mud flies off the tyre of a moving bicycle, in what direction does it fly? Explain.

vi Show that orbital angular momentum $L = mvr$

vii Define forced oscillations. Give its example.

viii If a mass spring system is hung vertically and set into oscillations, why does the motion eventually stop?

ix What happens to the period of a simple pendulum if its length is doubled?

x Define the terms node and antinode.

xi If a string vibrates in four segments at a frequency of 120 Hz, determine its fundamental frequency?

xii Is it possible for two identical waves travelling in the same direction along a string to give rise to a stationary wave? Explain.

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i How would you manage to get more orders of spectra using a diffraction grating?

ii Write two parts of Huygen's principle.

iii Write two conditions for detcctable interference.

iv If a person was looking through a telescope at the full moon, how would the appearance of moon be changed by covering half of the objective lens?

v What is optical fibre? Write down two uses of fibre optics.

vi Is it possible to convert internal energy into mechanical energy? Explain with an example.

vii Give an example of a process in which no heat is transferred to or from the system but the temperature of the system changes.

viii Derive Charles's law on the basis of kinetic molecular theory of gases.

ix Is it possible to construct a heat engine that will not expel heat into the atmosphere?

QSECTION - II. Note: Attempt any THREE questions.

Q5a. (Marks 5) Distinguish between elastic and inelastic collisions. Show that when two smooth balls undergo elastic collision in one dimension the magnitude of relative velocity of approach is equal to the magnitude of relative velocity of separation.

Q5b. (Marks 3) Two forces of magnitude 10 N and 20 N act on a in directions making angle 30° and 60° with x-axis respectively. Find the resultant force.

Q6a. (Marks 5) Prove that the work done in the gravitational field is independent of path followed by the body.

Q6b. (Marks 3) Find the temperature at which the velocity of sound in air is two times its velocity at 10°C.

Q7a. (Marks 5) Define terminal velocity. Show that terminal velocity is directly proportional to the square of radius.

Q7b. (Marks 3) A body of moment of inertia I = 0.80 kg m^2 about a fixed axis, rotates with a constant angular velocity of 100 rad s^{-1}. Calculate its angular momentum L and the torque to sustain this motion.

Q8a. (Marks 1,4) What is phase? Derive the formula of velocity in case of a horizontal mass-spring system.

Q8b. (Marks 3) A heat engine performs 100 J of work and at the same time ejects 400 J of heat energy to cold reservoirs. What is the efficiency of the engine?

Q9a. (Marks 5) What is compound microscope? Describe its construction and working. Also calculate its magnifying power.

Q9b. A light is incident normally on a grating which has 2500 lines per centimeter. Compute the wavelength of a spectral line for which the deviation in second order is 15°.

GROUP – II

Objective Type (Group-II)

Time Allowed: 20 Min. | **Maximum Marks:** 17

> Note: Inter, Part I, Lahore, 2021. Session 2017-2019 to 2020-2022. Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle in front of that question with Marker or Pen ink in the answer book. Cutting or filling two or more circles will result in zero mark in that question.

Q1. Venturi meter is a device used to measure:

(A) Density of fluid
(B) Speed of fluid
(C) Viscosity of fluid
(D) Pressure of fluid

Q2. The horizontal range of a projectile is:

(A) $2vi \sin\theta/g$ (B) $vi^2\sin^2\theta/2g$

(C) $vi^2 \sin2\theta/g$ (D) $vi \sin^2\theta/2g$

Q3. If 20 waves pass through the medium in 1 second with speed of $20ms^{-1}$, then the wavelength is:

(A) 20 m (B) 10 m

(C) 2 m (D) 1 m

Q4. The dimensions of torque are:

(A) $[MLT^{-2}]$ (B) $[ML^{-1}T^{-2}]$

(C) $[ML^{-1}T^{-1}]$ (D) $[ML^2T^{-2}]$

Q5. The efficiency of a carnot engine is:

(A) Infinite (B) Zero

(C) Greater than one (D) Less than one

Q6. One radian is equal to:

(A) 57.3° (B) 67.3°

(C) 87.3° (D) 60°

Q7. Which of the following is the only scalar quantity:

(A) Energy (B) Velocity

(C) Force (D) Torque

Q8. $[M^0L^0T^{-1}]$ refer to quantity:

(A) Velocity (B) Frequency

(C) Time period (D) Force

Q9. If length of the simple pendulum is double then its period increases:

(A) 1.44 times (B) 2 times

(C) 2.4 times (D) 3 times

Q10. For normal adjustment, length of telescope is:

(A) fo + fe (B) fo − fe

(C) fo/fe (D) fe/fo

Q11. The angular displacement per second is called angular:

(A) Acceleration (B) Rotation

(C) Velocity (D) Speed

Q12. SI unit of impulse is equivalent to:

(A) Force
(B) Momentum
(C) Torque
(D) Velocity

Q13. Pressure of an ideal gas can be written in terms of its density:

(A) $P = \rho\langle v^2 \rangle$
(B) $P = 1/3\ \rho\langle v^2 \rangle$
(C) $P = 2/3\ \rho\langle v^2 \rangle$
(D) $P = 1/2\ \rho\langle v^2 \rangle$

Q14. Kilowatt hour is the unit of:

(A) Power
(B) Work
(C) Force
(D) Momentum

Q15. Position vector of a point in x-z plane is given by:

(A) $y\hat{i} + z\hat{k}$
(B) $x\hat{i} + y\hat{k}$
(C) $x\hat{i} + z\hat{k}$
(D) $x\hat{i} + y\hat{j} + z\hat{k}$

Q16. The number of nodes between two consecutive antinodes is:

(A) 1
(B) 2
(C) 3
(D) Zero

Q17. Bragg's equation is given by:

(A) $d = 2\sin\theta/n\lambda$
(B) [illegible in source scan]
(C) $d = 2n\ \sin\theta/\lambda$
(D) $d = 2\lambda\ \sin\theta/n$

Essay Type (Group-II)

Time Allowed: 2.40 hrs. | **Maximum Marks:** 68

> Note: Session 2017-2019 to 2020-2022

QSECTION - I. (Marks 16)

QQ.2. (Marks 16) Write short answers to any EIGHT (8) questions:

i Define precision and accuracy.

ii Write down the two uses of dimensional analysis.

iii Calculate the volume of the cube and its uncertainty whose edge length is 2.25 ± 0.01 cm.

iv Write the dimensions of (a) Pressure (b) Density

v Suppose the sides of a closed polygon represent vectors arranged by head to tail. What is the sum of these vectors?

vi Show that the sum and difference of two perpendicular vectors of equal lengths are also perpendicular and of the same length.

vii If $A = 2i - 2j$, then what will be the orientation of A?

viii Define impulse and show that how it is related to linear momentum?

ix The horizontal range of projectile is four times of its height. What is the angle of projection of projectile?

x What is the principle of rocket propulsion?

xi Find the angle of projection of a projectile for which its height and range are equal.

xii Explain what do you understand by the term viscosity?

QQ.3. (Marks 16) Write short answers to any EIGHT (8) questions:

i State work energy principle, also write its relation.

ii An object has 1 J P.E. Explain what does it mean?

iii A girl drops a cup from a certain height which breaks into pieces. What energy changes are involved?

iv Find rotational K.E. of hoop.

v Define angular displacement. How its direction can be found?

vi Show that orbital angular momentum $L_o = mvr$

vii Explain the relation between total energy, potential energy and K.E. of a body oscillating with SHM.

viii If a mass spring system is hung vertically and set into oscillations, why does the motion eventually stop?

ix Does the acceleration of a simple harmonic oscillator remain constant during its motion? Is the acceleration ever zero? Explain.

x Describe the effect of density on speed of sound.

xi How are beats useful in tuning musical instruments?

xii Explain why sound travels faster in warm air than in cold air.

QQ.4. (Marks 12) Write short answers to any SIX (6) questions:

i Why the centre of Newton rings is dark?

ii Under what conditions two or more sources of light behave as coherent sources?

iii What is optical fibre? Write down two uses of fibre optics.

iv When the telescope is said to be in normal adjustment?

v Why would it be advantageous to use blue light with a compound microscope?

vi Define the term internal energy.

vii Prove that the maximum efficiency is always less than one or 100%.

viii Why does the pressure of a gas in a car tyre increases when it is driven through some distance?

ix Specific heat of a gas at constant pressure is greater than specific heat at constant volume. Why?

QSECTION - II. Note: Attempt any THREE questions.

Q5a. (Marks 5) Define scalar product of two vectors. Write down four characteristics of scalar product.

Q5b. (Marks 3) A proton moving with the speed of 1.0×10^7 ms^{-1} passes through a 0.020 cm thick sheet of paper, and emerges with a speed of 2.0×10^6 ms^{-1}. Assuming uniform deceleration. Find the retardation and time taken to pass through the paper.

Q6a. (Marks 5) What is variable force? Discuss work done by variable force.

Q6b. (Marks 3) The wavelength of signal from a radio transmitter is 1500 m and frequency is 200 kHz. What is wavelength for a transmitter operating at 1000 kHz and with what speed the radio waves travel?

Q7a. (Marks 5) Define centripetal force with an example. Prove that the magnitude of centripetal force is $F = m\omega^2$[illegible in source scan]

Q7b. (Marks 3) What gauge pressure is required in the city mains for a stream from a fire hose connected to the mains to reach a vertical height of 15m?

Q8a. Define molar specific heat of a gas and prove that $Cp - Cv = R$.

Q8b. (Marks 3) A spring whose spring constant is 80 Nm^{-1} vertically supports a mass of 1.0 kg in the rest position. Find the distance by which the mass must be pulled down, so that on being released it may pass the mean position with a velocity of 1.0 m/sec.

Q9a. (Marks 5) What is Astronomical Telescope? Describe its construction and working. Derive a formula to calculate its magnification power.

Q9b. (Marks 3) In a double slit experiment the second order maximum occurs at $\theta = 0.25°$. The wavelength is 650 nm. Determine the slit separation.